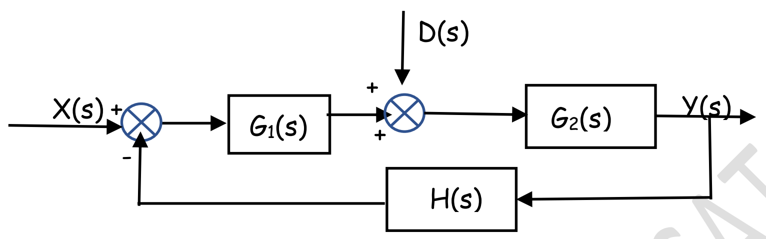
DISTURBANCE SIGNALS IN A FEEDBACK CONTROL SYSTEM

One of the important effect of feed back in a control system is the control and partial elimination of the effect of disturbance signals. A disturbance signal is an unwanted input signal that affects the system's output.

Consider a closed loop system. Let the plant is rpresented by $G_2(s)$ and the controller by $G_1(s)$.

A disturbance $D(s)$ acts on the system. Now, we have two inputs in to the system.



Since we are dealing with only linear systems, we can apply the principle of superposition. We will consider only one input at a time and find the corresponding output. The total output when both inputs are present will be equal to the sum of the individual outputs.

Let us assume $D(s) = 0$. The input is only $X(s)$.

The block diagram can be reduced to a single block of transfer function $T_1(s) = \frac{G_1 G_2}{1+G_1 G_2 H}$

$\frac{Y_1(s)}{X(s)} = \frac{G_1 G_2}{1+G_1 G_2 H}$ from which $Y_1(s) = X(s)\left(\frac{G_1 G_2}{1+G_1 G_2 H}\right)$

Now consider that $X(s) = 0$. The input is only $D(s)$. Now the blocks in feedback path becomes $(-H)$ and (G_1)

$T_2(s) = \frac{G_2}{1+G_1 G_2 H}$

$\frac{Y_2(s)}{D(s)} = \frac{G_2}{1+G_1 G_2 H}$ from which $Y_2(s) = D(s)\left(\frac{G_2}{1+G_1 G_2 H}\right)$

When both inputs are present, $Y(s) = Y_1(s) + Y_2(s)$

$$Y(s) = X(s)\left(\frac{G_1 G_2}{1+G_1 G_2 H}\right) + D(s)\left(\frac{G_2}{1+G_1 G_2 H}\right) = \left(\frac{G_2}{1+G_1 G_2 H}\right)\{G_1 X(s) + D(s)\}$$

Consider the case $|G_1 H| \gg 1$ and $|G_1 G_2 H| \gg 1$ (Modulii of the corresponding complex variable)

$Y_2(s) = D(s)\left(\frac{G_2}{1+G_1 G_2 H}\right) \cong D(s)\frac{G_2}{G_1 G_2 H} \cong D(s)\frac{1}{G_1 H} \cong 0$ (As $|G_1 H| \gg 1$, $\frac{1}{G_1 H} \cong 0$)

That is, the effect of the disturbance is suppressed. This is an advantage of feedback systems.

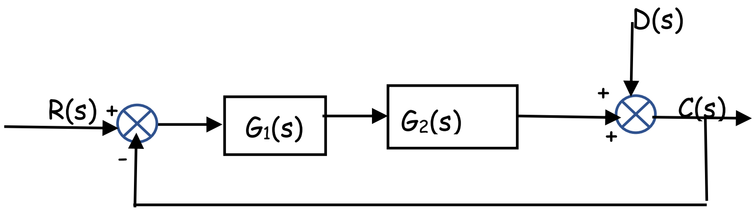
Now, $Y_1(s) = X(s)\left(\frac{G_1 G_2}{1+G_1 G_2 H}\right) \cong X(s)\frac{G_1 G_2}{G_1 G_2 H} \cong X(s)\frac{1}{H}$

This means that if $|G_1 G_2 H| \gg 1$, then the closed loop transfer function $\frac{Y_1(s)}{X(s)}$ becomes independent of $G_1(s)$ and $G_2(s)$ and becomes inversely proportional to $H(s)$. This is another advantage of closed loop system. If we have a unity feed back system where $H(s) = 1$, then the actual ouput becomes equal to the input (which is the desired output).

Exercise 20

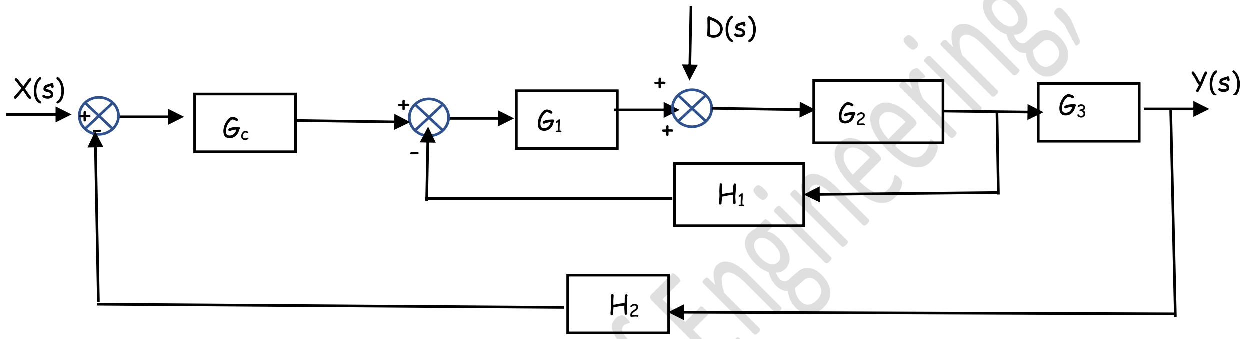
20.1

The figure shows a closed loop system with reference input R(s) and disturbance input D(s). Obtain the expression for the output C(s) when both are present.



20.2

Obtain the transfer function Y(s)/X(s) and Y(s)/D(s) for the system shown below.



20.3

Obtain the expression for steady state error for the system shown if both R(s) and D(s) are present.